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Course/Section: CPE212 - CPE31S1	**Date Submitted: 02/19/2026**
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Activity 3: Install SSH server on CentOS or RHEL 8

1. **Objectives:**

1.1 Install Community Enterprise OS or Red Hat Linux OS

1.2 Configure remote SSH connection from remote computer to CentOS/RHEL-8

2. **Discussion:**

CentOS vs. Debian: Overview

CentOS and Debian are Linux distributions that spawn from opposite ends of the candle.

CentOS is a free downstream rebuild of the commercial Red Hat Enterprise Linux distribution where, in contrast, Debian is the free upstream distribution that is the base for other distributions, including the Ubuntu Linux distribution.

As with many Linux distributions, CentOS and Debian are generally more alike than different; it isn't until we dig a little deeper that we find where they branch.

CentOS vs. Debian: Architecture

The available supported architectures can be the determining factor as to whether a distro is a viable option or not. Debian and CentOS are both very popular for x86_64/AMD64, but what other archs are supported by each?

Both Debian and CentOS support AArch64/ARM64, armhf/armhfp , i386 , ppc64el/ppc64le. (Note: armhf/armhfp and i386 are supported in CentOS 7 only.)

CentOS 7 additionally supports POWER9 while Debian and CentOS 8 do not. CentOS 7 focuses on the x86_64/AMD64 architecture with the other archs released through the AltArch SIG (Alternate Architecture Special Interest Group) with CentOS 8 supporting x86_64/AMD64, AArch64 and ppc64le equally.

Debian supports MIPSel, MIPS64el and s390x while CentOS does not. Much like CentOS 8, Debian does not favor one arch over another —all supported architectures are supported equally.

CentOS vs. Debian: Package Management

Most Linux distributions have some form of package manager nowadays, with some more complex and feature-rich than others.

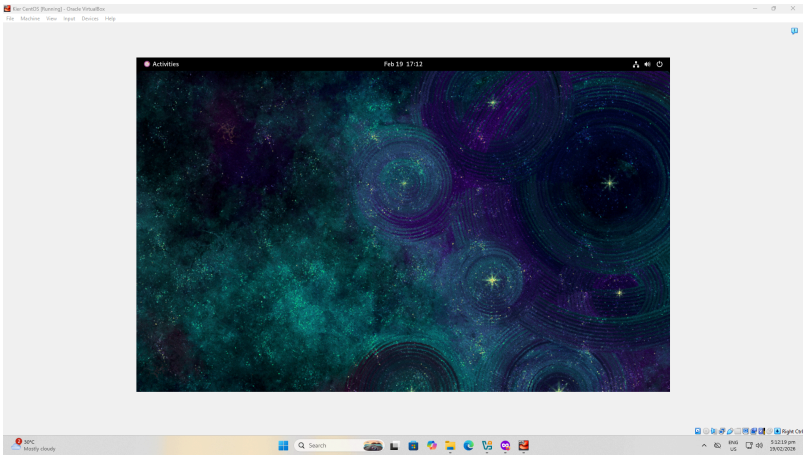
CentOS uses the RPM package format and YUM/DNF as the package manager.

Debian uses the DEB package format and dpkg/APT as the package manager.

Both offer full-feature package management with network-based repository support, dependency checking and resolution, etc.. If you're familiar with one but not the other, you may have a little trouble switching over, but they're not overwhelmingly different. They both have similar features, just available through a different interface.

Task 1: Download the CentOS or RHEL-8 image (Create screenshots of the following)

1. Download the image of the CentOS here: http://mirror.rise.ph/centos/7.9.2009/isos/x86_64/
2. Create a VM machine with 2 Gb RAM and 20 Gb HD.
3. Install the downloaded image.
4. Show evidence that the OS was installed already.



Task 2: Install the SSH server package *openssh*

1. Install the ssh server package *openssh* by using the *dnf* command:
   $ dnf install openssh-server

```
[root@vbox kier]# dnf install openssh-server
Updating Subscription Management repositories.
Unable to read consumer identity

This system is not registered with an entitlement server. You can use "rhc" or "
subscription-manager" to register.

CentOS Stream 9 - BaseOS                        1.3 MB/s | 8.9 MB     00:06
CentOS Stream 9 - AppStream                     1.6 MB/s |  27 MB     00:16
CentOS Stream 9 - Extras packages                17 kB/s |  20 kB     00:01
Package openssh-server-8.7p1-43.el9.x86_64 is already installed.
Dependencies resolved.
================================================================================
 Package                Architecture  Version                Repository     Size
================================================================================
Upgrading:
 openssh                x86_64        9.9p1-3.el9            baseos        424 k
 openssh-clients        x86_64        9.9p1-3.el9            baseos        771 k
 openssh-server         x86_64        9.9p1-3.el9            baseos        544 k

Transaction Summary
================================================================================
Upgrade  3 Packages

Total download size: 1.7 M
Is this ok [y/N]: Y
Downloading Packages:
(1/3): openssh-server-9.9p1-3.el9.x86_64.rpm    564 kB/s | 544 kB     00:00
(2/3): openssh-9.9p1-3.el9.x86_64.rpm           420 kB/s | 424 kB     00:01
(3/3): openssh-clients-9.9p1-3.el9.x86_64.rpm   703 kB/s | 771 kB     00:01
--------------------------------------------------------------------------------
Total                                           1.1 MB/s | 1.7 MB     00:01
CentOS Stream 9 - BaseOS                        1.6 MB/s | 1.6 kB     00:00
Importing GPG key 0x8483C65D:
 Userid     : "CentOS (CentOS Official Signing Key) <security@centos.org>"
 Fingerprint: 99DB 70FA E1D7 CE22 7FB6 4882 05B5 55B3 8483 C65D
 From       : /etc/pki/rpm-gpg/RPM-GPG-KEY-centosofficial
```

2. Start the *sshd* daemon and set to start after reboot:
   $ systemctl start sshd
   $ systemctl enable sshd

```
[root@vbox kier]# systemctl start sshd
[root@vbox kier]# systemctl enable sshd
```

3. Confirm that the sshd daemon is up and running:
   $ systemctl status sshd

```
[root@vbox kier]# systemctl status sshd
● sshd.service - OpenSSH server daemon
     Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: ena>
     Active: active (running) since Thu 2026-02-19 17:04:42 PST; 2min 57s ago
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 6827 (sshd)
      Tasks: 1 (limit: 60072)
     Memory: 1.1M
        CPU: 6ms
     CGroup: /system.slice/sshd.service
             └─6827 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Feb 19 17:04:42 vbox systemd[1]: Starting OpenSSH server daemon...
Feb 19 17:04:42 vbox sshd[6827]: Server listening on 0.0.0.0 port 22.
Feb 19 17:04:42 vbox sshd[6827]: Server listening on :: port 22.
Feb 19 17:04:42 vbox systemd[1]: Started OpenSSH server daemon.
lines 1-16/16 (END)
```

4. Open the SSH port 22 to allow incoming traffic:

$ firewall-cmd --zone=public --permanent --add-service=ssh
$ firewall-cmd --reload

```
[root@vbox kier]# firewall-cmd --zone=public --permanent --add-service=ssh
Warning: ALREADY_ENABLED: ssh
success
[root@vbox kier]# firewall-cmd --reload
success
```

5. Locate the ssh server man config file /*etc/ssh/sshd_config* and perform custom configuration. Every time you make any change to the /*etc/ssh/sshd-config* configuration file reload the *sshd* service to apply changes:
$ systemctl reload sshd

```
[root@vbox kier]# systemctl reload sshd
```

Task 3: Copy the Public Key to CentOS

1. Make sure that *ssh* is installed on the local machine.
2. Using the command *ssh-copy-id*, connect your local machine to CentOS.

```
kier@LocalMachine:~$ ssh-copy-id kier@192.168.56.114
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/kier/.ssh/i
d_ed25519.pub"
The authenticity of host '192.168.56.114 (192.168.56.114)' can't be established.
ED25519 key fingerprint is SHA256:wYCZFrMgAm3T50iLjiwlNYo25qTDeAGCxJDlRMjhD3c.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
The authenticity of host '192.168.56.114 (192.168.56.114)' can't be established.
ED25519 key fingerprint is SHA256:wYCZFrMgAm3T50iLjiwlNYo25qTDeAGCxJDlRMjhD3c.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompt
ed now it is to install the new keys
kier@192.168.56.114's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'kier@192.168.56.114'"
and check to make sure that only the key(s) you wanted were added.
```

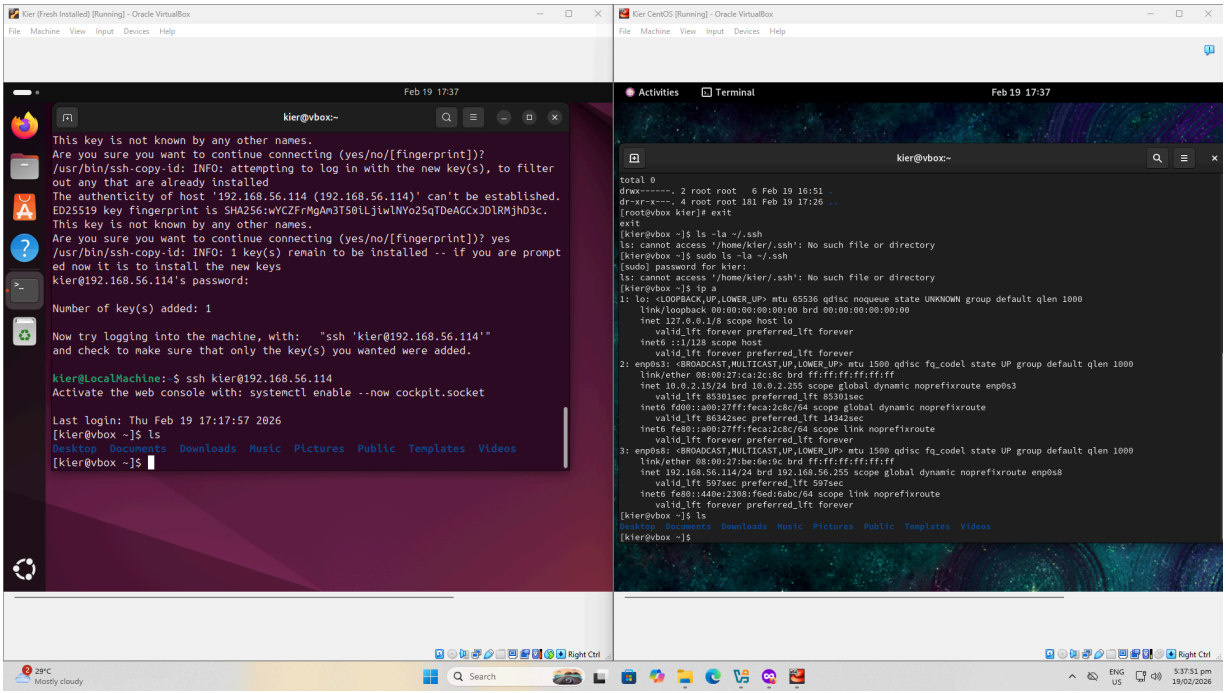
3. On CentOS, verify that you have the *authorized_keys*.

```
[kier@vbox ~]$ ls -la ~/.ssh
total 8
drwx------.  2 kier kier   29 Feb 19 17:36 .
drwx------. 15 kier kier 4096 Feb 19 17:36 ..
-rw-------.  1 kier kier   99 Feb 19 17:36 authorized_keys
```

Task 4: Verify ssh remote connection

1. Using your local machine, connect to CentOS using ssh.

2. Show evidence that you are connected.



Reflections:

Answer the following:

1. What do you think we should look for in choosing the best distribution between Debian and Red Hat Linux distributions?
   - When choosing between Debian and Red Hat, I think we should look at things like how stable the system is, what kind of support it offers (community vs. enterprise), and what we plan to use it for, like personal projects or business environments. It also depends on which package manager we prefer and how long we need support for updates.
2. What are the main differences between Debian and Red Hat Linux distributions?
   - The main difference is that Debian is community-based and uses APT with .deb packages, while Red Hat is backed by a company and uses DNF/YUM with .rpm packages. Red Hat is usually more focused on enterprise and professional support, while Debian is known for being very stable and free to use.